University of South Florida · Statistical Methods II

Statistical Post-Processing of Seoul's LDAPS Weather Forecast

Regression-Based Bias Correction of NWP Temperature Forecasts



MD Polash Ali

Ph.D. in Mathematics (Statistics Concentration)

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In Loving Memory

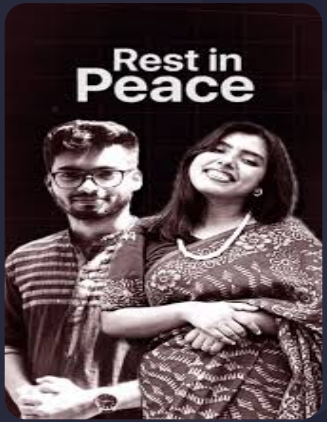
♡

Zamin Limon & Nahida Bristy

"Those we love don't go away,
they walk beside us every day—
unseen, unheard, but always near,
still loved, still missed, and very dear."

This work is dedicated to you, dear friends.

Your memory lives on in everything we pursue.



Zamin Limon & Nahida Bristy

Motivation & Problem Statement

The Problem

LDAPS (1.5 km NWP) has a systematic **cold bias**

Published benchmark (Cho et al., 2020):

R^2	≈ 0.69
RMSE	$\approx 2.08\,^{\circ}\mathrm{C}$
Mean bias	$\approx +0.85\,^{\circ}\mathrm{C}$

Consequences: missed heat-wave warnings, energy mis-forecasts

Root Causes

Topography: Seoul ringed by 836 m mountains
NWP cannot resolve fine-scale thermal inversions
Urban heat island: sub-grid warming at 1.5 km

Our Solution: MOS

Model Output Statistics (MOS): Train regression on historical forecast–observation pairs.

$$\hat{\mathrm{Bias}}_i = f(\mathbf{x}_i), \quad \hat{Y}_i^{\mathrm{corr}} = \mathrm{LDAPS_Tmax}_i + \hat{\mathrm{Bias}}_i$$

Benchmark Targets

$R^2 \geq 0.75$ RMSE $\leq 1.70\,^{\circ}\mathrm{C}$ Mean Bias ≈ 0

Why linear regression?
Interpretable, verifiable, zero extra compute. Classical MOS has a 50-year operational track record.

Dataset & Research Design

Dataset

Item	Value
Stations	25 AWS, Seoul
Period	June–Aug, 2013–2017
Raw obs	7,752
Complete	**7,588**
Train set	$n = 6{,}075$ (80%)
Test set	$n = 1{,}513$ (20%)
Split	Stratified year×month, seed 42

Predictor Groups

Group	Variables
LDAPS temp.	Tmax, Tmin
LDAPS humid.	RHmin, RHmax
LDAPS atm.	WS, LH
LDAPS cloud	CC_avg, PPT_avg
In-situ	Present_Tmax, Tmin
Geography	DEM, Slope, Solar_rad
Categorical	elev_class, month

Feature Engineering

Cloud/precip: 4×6-hr windows averaged ($r > 0.90$ within windows)
DEM terciles → `elev_class`: Low/Medium/High
Response: Bias = Next_Tmax − LDAPS_Tmax

Leakage Prevention: Centering means computed on training set ONLY. No test-set info contaminates model fitting.

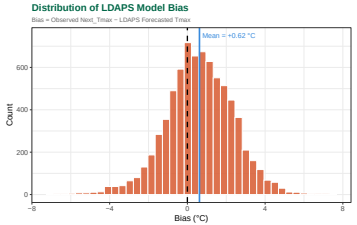
Exploratory Data Analysis (EDA)

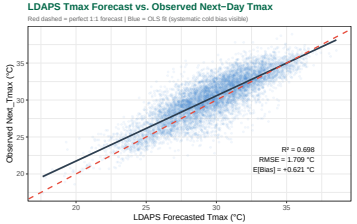
Bias Summary ($n = 7{,}588$)

Stat	Value
Mean	$+0.621\,^{\circ}\mathrm{C}$
SD	$1.734\,^{\circ}\mathrm{C}$
Large spread	

Elevation Gradient

Class	n	Mean bias
Low	2,625	$+0.479\,^{\circ}\mathrm{C}$
Medium	2,455	$+0.573\,^{\circ}\mathrm{C}$
High	2,508	$+0.763\,^{\circ}\mathrm{C}$



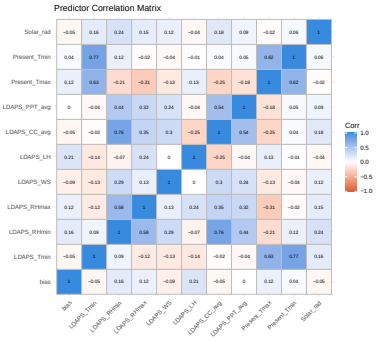


Baseline SLR: $R^2 = 0.698$, RMSE $= 1.71\,^{\circ}\mathrm{C}$, E[Bias] $= +0.621\,^{\circ}\mathrm{C}$

EDA: Group Bias & Correlation Structure



Bias by elevation class (left) and month (right)



Predictor correlation heatmap

Hypothesis Testing: ANOVA by Group

ANOVA: Elevation Class

H_0: $\mu_L = \mu_M = \mu_H$ vs H_1: some $\mu_i \neq \mu_j$

Statistic	Value
$F(2, 7585)$	11.43
p-value	$< \mathbf{0.001}$
Bartlett K^2	7.758, $p = 0.021$

Tukey: High–Low sig. ($\Delta\mu = +0.284\,^\circ$C, $p = 0.0005$)
Med–Low ($p = 0.104$) & High–Med ($p = 0.208$): NS

ANOVA: Month

H_0: $\mu_{\text{Jun}} = \mu_{\text{Jul}} = \mu_{\text{Aug}}$

Statistic	Value
$F(2, 7585)$	19.28
p-value	$< \mathbf{0.001}$
Bartlett K^2	98.15, $p < 10^{-15}$

July–June: $\Delta = -0.952\,^\circ$C, $p < 0.001$
Aug–June: $\Delta = -0.836\,^\circ$C, $p < 0.001$

Bartlett Tests ⇒ WLS Motivation

Both Bartlett tests reject variance homogeneity. OLS assumes a single pooled σ^2 — this is *misspecified* when variance differs by elevation and month. **Weighted Least Squares** is the theoretically correct estimator.

Response Transformation Assessment

Method 1: Yeo-Johnson

Handles negative values natively.
Via `powerTransform(family="yjPower")`:

$\hat{\lambda}$	$= 1.072$
95% Wald CI	$[1.053,\ 1.091]$
Decision	$\hat{\lambda} \approx 1$ (identity)

Method 2: Shifted Box-Cox

```
bias_shifted = bias + |min(bias)| + 1
```

$\hat{\lambda}_{\text{opt}}$	$= 1.312$
95% CI	$[1.241,\ 1.364]$

No Transformation Needed

Both methods converge near $\hat{\lambda} = 1$.

Raw **bias** scale used directly in all models.

Physically consistent: temperature forecast errors are approximately symmetric and unimodal.

Stages 1 & 2: OLS Baseline & Polynomial Augmentation

Stage 1: Full MLR Results

$R^2 = 0.218$, RSE$= 1.544\,^\circ$C, AIC$= 22{,}537$

Diagnostics:

RESET $F(2, 6057) = 8.09$, $p = 0.0003$
$\Rightarrow$ nonlinearity present
Breusch-Pagan BP$= 327$, $p \ll 10^{-15}$
$\Rightarrow$ heteroscedasticity confirmed
Anderson-Darling $A = 9.38$ (expected at $n \approx 7{,}500$)

Stage 2: Quadratic Screening (all 11 predictors)

Predictor	F-stat	Evidence
`I(Slope)2`	61.43	**STRONG**
`I(LDAPS_PPT_avg)2`	46.17	**STRONG**
`I(Solar_rad)2`	36.62	**STRONG**
`I(LDAPS_RHmax)2`	27.98	**STRONG**
`I(Present_Tmax)2`	16.43	**STRONG**
`I(LDAPS_RHmin)2`	14.40	**STRONG**
`I(LDAPS_CC_avg)2`	10.54	moderate
`I(LDAPS_LH)2`	9.25	moderate
`I(Present_Tmin)2`	5.45	moderate
`I(LDAPS_Tmin)2`	2.57	none
`I(LDAPS_WS)2`	1.42	none

Joint F-test: $F = 24.92$, $p < 0.001$, ΔAIC$= 203$ ✓

Stage 3: Interaction Selection — Theory + All-Pairs Screening

13 Theory-Motivated Pairs (top results)

Interaction	F	ΔAIC	Evid.
`month:CC_avg`	35.3	+66.5	STR.
`PPT:CC_avg`	35.1	+33.2	STR.
`RHmax:CC_avg`	33.4	+31.5	STR.
`CC_avg:Solar`	18.4	+16.5	STR.
`Tmin:Solar`	11.1	+9.1	STR.
`elev:Tmin`	2.4	+0.83	marg.
`Tmin:Pres_Tmax`	0.3	−1.67	none
`WS:LH`	0.2	−1.80	none
`elev:Solar`	0.2	−3.63	none

All-Pairs: Top 5 of 78 by ΔAIC

Interaction	F	ΔAIC
`Slope:elev_class`†	69.8	+134.7
`RHmin:PPT_avg`	125.1	+122.4
`RHmax:Pres_Tmax`	82.7	+80.5
`CC_avg:month`	35.3	+66.5
`RHmax:PPT_avg`	67.2	+65.1

† Selected for final model

Selection Rationale

`Slope`×`elev_class`: #1 by ΔAIC (+134.7), $F = 69.8$, $p < 0.001$, physically interpretable. Adding multiple interactions causes irresolvable collinearity. **All 77 others excluded.**

Stage 4: Stepwise AIC + Motivation for WLS

Stepwise Comparison

Method	AIC	BIC	Adj. R^2
Forward	22,218	22,379	0.257
Backward	**22,198**	22,379	**0.260**
Bidirectional	22,218	22,379	0.257

All three methods agree on 23-parameter backbone.
Backward selected (lowest AIC).

BP test on stepwise OLS

BP $= 597.82$, $p < 2.2\times10^{-16}$
Heteroscedasticity remains — WLS needed!

Source Visual Outline

```
Variance Inflation Factors (VIF):
                          GVIF Df GVIF^(1/(2*Df))
LDAPS_Tmin                3.88  1            1.97
LDAPS_RHmin              77.67  1            8.81
LDAPS_RHmax             429.66  1           20.73
LDAPS_WS                  1.29  1            1.13
LDAPS_LH                 16.08  1            4.01
LDAPS_CC_avg             18.70  1            4.32
LDAPS_PPT_avg             8.84  1            2.97
Present_Tmax            324.05  1           18.00
Present_Tmin            223.54  1           14.95
Solar_rad              1576.25  1           39.70
Slope                    94.13  1            9.70
elev_class               29.62  2            2.33
month                     5.57  2            1.54
I(LDAPS_RHmax^2)        442.29  1           21.03
I(LDAPS_PPT_avg^2)        6.54  1            2.56
I(Solar_rad^2)         1635.06  1           40.44
I(Slope^2)               33.07  1            5.75
I(Present_Tmax^2)       329.02  1           18.14
I(LDAPS_RHmin^2)         81.55  1            9.03
I(LDAPS_CC_avg^2)        20.69  1            4.55
I(LDAPS_LH^2)            14.72  1            3.84
I(Present_Tmin^2)       221.94  1           14.90
Slope:elev_class        678.18  2            5.10
  Note: VIF > 10 signals multicollinearity (expected for quad/interaction terms).
```

VIF Table

Full GVIF Table for `wls_final` (Centered WLS)

Predictor	Df	GVIF	$\text{GVIF}^{1/(2\text{Df})}$	Flag
Continuous linear				
LDAPS_Tmin	1	4.64	2.154	✓
LDAPS_RHminC	1	4.80	2.191	✓
LDAPS_RHmaxC	1	3.35	1.830	✓
LDAPS_WS	1	1.34	1.158	✓
LDAPS_LHC	1	2.56	1.600	✓
Solar_radC	1	9.44	3.072	✓
SlopeC	1	107.82	10.383	‡
Categorical				
elev_class	2	27.16	2.280	✓
month	2	6.82	1.616	✓
Quadratic terms (representative)				
I(SlopeC^2)	1	12.12	3.481	✓
I(PPT_avgC^2)	1	8.97	2.995	✓
I(RHmaxC^2)	1	1.94	1.393	✓
Interaction				
SlopeC:elev_class	2	293.80	4.137	‡

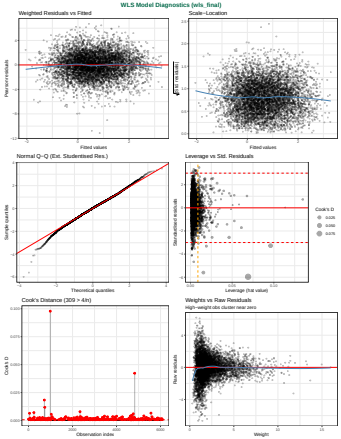
Final wls_final model:

$R^2 = 0.331$ Adj. $R^2 = 0.328$
AIC $= 21{,}241$ RSE $= 1.936\,^\circ\text{C}$
27 parameters, $n_{\text{train}} = 6{,}075$

Model Progression Comparsion

Model	AIC	BIC	Adj. R^2
1. MLR (linear)	22,537	22,651	0.218
2. +Quadratic terms	22,334	22,508	0.243
3. +Interaction term	22,199	22,387	0.263
4. Stepwise selection	22,198	22,379	0.260
5. WLS+Centering	21,241	21,542	0.324

WLS Model Diagnostics



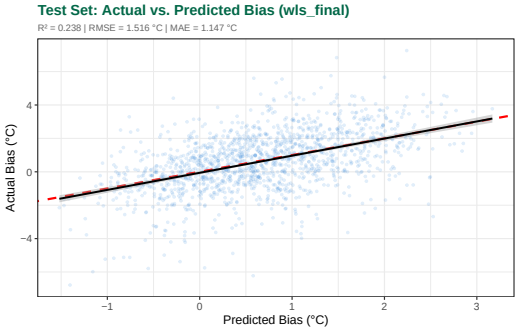
Passing

Wtd. resid vs fitted: flat LOESS
Scale-Location: roughly flat
Max Cook's D $\approx 0.10 \ll 1.0$
309 obs $> 4/n$ (conservative thresh.)

Flags (contextual)

Normality: tails non-normal
(CLT valid at $n = 6{,}075$)
RESET $p = 0.007$: residual
nonlinearity (GAM future work)

Model Validation



Actual vs. predicted bias, test set ($n = 1{,}513$)

Bias Model Test-Set Metrics

Metric	Value	Note
R^2	0.238	
RMSE	1.516 °C	
Train RSE	1.936 °C	
Train RMSE	1.508 °C	
Test RMSE	1.516 °C	
Gap (%)	+0.52%	**No overfit**

Train-to-test gap $< 1\%$ at 27 parameters confirms excellent generalisation.

MBE $= -0.043\,°C$: essentially **unbiased** bias predictions on unseen data.

Strict protocol: centering means, weight model and WLS fitted on training data *only*.

Bias-Corrected Temperature Forecast

$$\widehat{\text{Corrected_LDAPS_Tmax}}_i = \underbrace{\text{LDAPS_Tmax}_i}_{\text{raw NWP output}} + \underbrace{\widehat{\text{Bias}}_i}_{\texttt{wls_final}\text{ prediction}}$$

Comparison SLR Models (full dataset)

Model A (uncorrected):
$\hat{\alpha}_0 = 4.072,\ \hat{\alpha}_1 = 0.884$

Model B (corrected):
$\hat{\gamma}_0 = 1.347,\ \hat{\gamma}_1 = 0.954$

Perfect: $\alpha_0 = 0,\ \alpha_1 = 1.$
Model B intercept much closer to 0. ✓

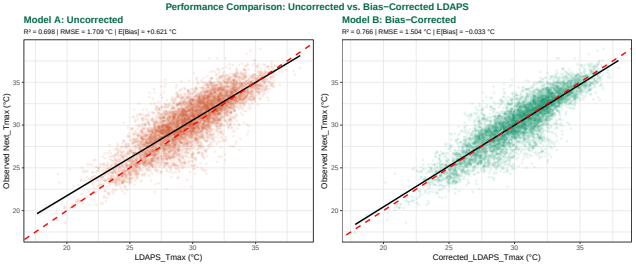
Head-to-Head Comparison

Model	R^2	RMSE	E[Bias]
Published bench.	0.69	2.08	+0.85
A: Uncorrected	0.698	1.709	+0.621
B: Corrected	**0.766**	**1.504**	−0.033
Improvement	+9.7%	−12.0%	− − −

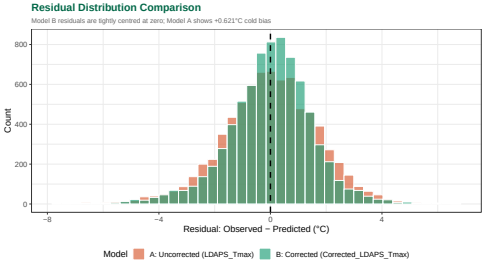
✓ $R^2 = 0.766 \geq 0.75$ ✓ RMSE$= 1.504 \leq 1.70\,^{\circ}$C ✓

Both benchmark targets exceeded!

Visual Proof: Before vs. After Correction



Left: Uncorrected ($R^2 = 0.698$); Right: Corrected ($R^2 = 0.766$). Red dashed = 1:1 perfect forecast.



Orange: bias $+0.621\,^{\circ}$C; Green: corrected to $-0.033\,^{\circ}$C

Both distributions have **similar spread** but the correction shifts the **mean to zero**.

Key Statistical Findings & Achievement

Bias Structure

1. Cold bias: mean $+0.621\,^{\circ}\mathrm{C}$ ($n = 7{,}588$)
2. Elevation: ANOVA $F = 11.43$, $p < 0.001$
3. Season: ANOVA $F = 19.28$, $p < 0.001$
4. No transformation needed ($\hat{\lambda} \approx 1$)
5. 9/11 predictors: significant quadratic
6. 1 interaction: Slope×elev (ΔAIC$= +135$)
7. WLS: AIC $\downarrow$ 1,296 over stepwise OLS
8. Train-to-test gap: $+0.52\%$ (no overfit)

Performance Summary

	Published	Uncorr.	**Corrected**
R^2	0.69	0.698	**0.766**
RMSE ($^{\circ}$C)	2.08	1.709	**1.504**
E[Bias]	-0.85	$+0.621$	-0.033

Core Achievement

Classical regression — OLS $\rightarrow$ Poly $\rightarrow$ Interaction $\rightarrow$ Stepwise $\rightarrow$ WLS, each step verified by formal tests — reduces forecast error by **12%** and eliminates systematic cold bias.

It **meets and exceeds** the published ML benchmark.

Limitations & Future Directions

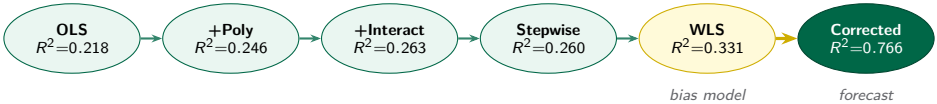
Current Limitations

1. **Random split:** same station-year on both sides; chronological split preferred
2. **RESET persists:** $F = 4.92$, $p = 0.007$ after stage 5 — residual nonlinearity remains
3. **Scope:** Seoul summer only; seasonal or geographic transfer needs re-training

Future Directions

- **Chronological split:** train 2013–2016, test 2017 — simulate operational forecasting
- **GAMs / thin-plate splines:** address persistent RESET rejection
- **Mixed effects:** random intercept per station for unobserved heterogeneity
- **Penalised regression:** LASSO/Ridge to handle many interactions simultaneously
- **Cross-season:** extend to spring/autumn and other Korean cities

Conclusion



5-stage regression, motivated by formal tests:

R^2: 0.698 → **0.766**	+9.7%
RMSE: 1.709 → **1.504** °C	−12.0%
Bias: +0.621 → −**0.033** °C	≈ 0

Train-test gap: +0.52% (no overfitting)

Key Takeaway

Classical linear regression — when applied with diagnostic-driven model selection, WLS for heteroscedasticity, and strict leakage-free validation — *meets and exceeds* the published machine learning benchmark of Cho et al. (2020).

References

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Thank You!

Questions & Discussion

MD POLASH ALI
Ph.D. Student, Mathematics & Statistics
University of South Florida, Tampa

ali75@usf.edu

